Files needed for Simulated Data results -

DD-STScan Data Simulation.R - Provides the code for simulating data under 36 cluster models.

Data_pop_summary.csv / adj_matrix.csv / New.precog.sim.R / New.mst.all.R - Files needed to run DD-STScan Data Simulation.R.

DD-STScan.R - Provides the code for the DD-STScan method.

Cylindrical/Elliptical.ipynb - Provides the code for running iterations using the cylindrical and elliptical methods.

Prm4.tex - Provides the parameter file used for the cylindrical and elliptical methods in `Cylindrical/Elliptical methods.ipynb`. `SpatialWindowShapeType` was set to 0 for the cylindrical method and updated to 1 when changing to the elliptical method.

NMgeo.csv - Provides the geometric data needed to run the `Cylindrical/Elliptical methods.ipynb`.

Simulated Data folder - Provides the set of datasets simulated using `DD-STScan Data Simulation.R`.

Results for Simulated Data.pdf - "Sen" denotes sensitivity. The reported sensitivity, PPV, and F1 scores are the average values obtained across 1,000 datasets simulated under each cluster model.

Real world application -

Data source - https://www.cdc.gov/united-states-cancer-statistics/public-use/index.html

The real-world data analysis was conducted using the same DD-STScan code and the corresponding implementations of the cylindrical and elliptical scan statistics used in the simulation studies.

Empirical Type I Error under the Null Hypothesis -

The same simulation and method codes described above were used for the empirical Type I error analysis. For these simulations, no spatio-temporal units were designated as clusters, representing the null hypothesis. The resulting datasets were analyzed using the DD-STScan,

cylindrical, and elliptical methods, and the empirical Type I error was calculated from the resulting analyses.

Simulated Null Datasets folder - Provides the datasets simulated under the Null hypothesis.

Null Data Results.csv - Provides the empirical Type I error results obtained from the simulated datasets.

State_coordinates.csv - Is the file with geographical data used in Cylindrical/Elliptical.ipynb.

Software requirements:

- R / RStudio
- Python /Jupyter Notebook or Google Colab
- SaTScan v10.3.2

R Packages: Smerc / scanstatistics

Python Packages: Pandas